

Taiwanese Hokkien in AI: Challenges, Approaches, and Language Modeling

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Abstract—In the evolving landscape of artificial intelligence (AI) and natural language processing (NLP), major advancements have been made, particularly with powerful pre-trained language models. However, these models often overlook region-specific languages like Taiwanese Hokkien. This paper addresses the challenges in integrating Taiwanese Hokkien into NLP and proposes solutions. We employ deep neural networks to develop a translation model that unifies diverse Taiwanese Hokkien texts into a standardized script. Leveraging this unified text, we train pre-trained language and generative models tailored for Taiwanese Hokkien. These models are pivotal in facilitating the integration of Taiwanese Hokkien into the AI landscape, enabling exploration of its various aspects in NLP. Our experiments demonstrate the successful acquisition of a universal language representation by the pre-trained language model and the impressive generation of fluent Taiwanese Hokkien text by the generative model.

Keywords—Taiwanese Hokkien, natural language processing, Transformer architecture, BERT, GPT

I. INTRODUCTION

In 1964, Professor Joseph Weizenbaum introduced Eliza, the pioneering chatbot that marked the inception of conversational AI, showcasing its conversational prowess and passing the Turing test [1]. AI achieved a significant breakthrough in 2012 with deep convolutional neural networks (CNNs) for image recognition, introducing techniques like dropout and ReLU to enhance model performance [2].

In 2016, AlphaGo defeated the world champion in Go, highlighting AI's potential, though progress in natural language processing (NLP) remained limited [3]. In November 2022, ChatGPT emerged as a transformative AI model, demonstrating remarkable abilities in various tasks.

This paper addresses the limited attention given to regional languages like Taiwanese Hokkien in NLP development, leveraging state-of-the-art AI models to develop language models specifically for it. Using Transformer architecture, we train translation models and develop a unified script system. Subsequently, we train a universal language model (Hokkien-BERT) and a generative model (Hokkien-GPT) tailored for Taiwanese Hokkien [4].

The paper includes background on Taiwanese Hokkien, analysis of training data, model architectures, research outcomes, and conclusions with future research directions.

II. REASEARCH BACKGROUND

In this section, we'll explore the historical evolution and current status of Mainland China's Minnan dialect and

Taiwanese Hokkien, along with their associated writing systems.

A. Minnan Dialect: A Brief Overview

Originating from Quanzhou, China, the Minnan dialect is spoken widely in southeastern coastal areas, encompassing regions of Fujian, Guangdong, Taiwan, Singapore, Malaysia, Indonesia, and Vietnam. With approximately 49.7 million speakers globally, it ranks as the 37th most spoken language worldwide [5].

B. Taiwanese Hokkien: History and Usage

Taiwanese Hokkien, a branch of the Minnan dialect, traces its origins back to various Chinese immigrant waves, including migrations during the Dutch colonial period, the late Ming Dynasty, and the Qing Dynasty's rule. Over time, it has evolved into one of the major languages of Taiwan, with distinct regional variations and influences from indigenous languages and foreign powers like Japan.

C. Writing Systems of Taiwanese Hokkien

Taiwanese Hokkien employs four main writing systems: Full-han, Church Romanization, Han-lo, and Tai-lo. Each system has its historical context and usage, reflecting the linguistic diversity and cultural heritage of Taiwanese Hokkien [6].

III. ANAYLYSIS OF TRAINING CORPORA

Our training data consists of two categories: one for translation models, focusing on short sentences and vocabulary, and another for language models (LMs), comprising longer texts from various online sources. Compiling corpora for Taiwanese Hokkien has faced significant challenges, impacting its progress in natural language processing (NLP). In the following subsections, we present our corpora sources, discuss encountered challenges, and propose solutions.

A. Data Sources

For translation models, we use the "Dictionary of Frequently-Used Taiwan Minnan" website, offering approximately 24,000 Taiwanese vocabulary entries [7]. The LM training corpus consists of longer texts from diverse online sources, totaling around 43 million Chinese characters.

B. Challenges and Solutions

During corpus collection, we faced critical challenges:

- **Non-unified Writing Systems:** Inconsistencies arise due to Taiwanese Hokkien employing four writing systems.

- **Data Sparsity:** Limited literary creation in Taiwanese Hokkien leads to sparse corpora availability.
- **Consistency Issues in Unicode Encoding:** Encoding differences in writing systems pose challenges in data processing.

C. Solutions

To address data sparsity, we employ data augmentation techniques and develop a translation workflow to unify writing systems and ensure consistent Unicode encoding for collected materials.

For data augmentation, We train a Taiwan Mandarin-to-Full-han translation model and use word-pair data for Transformer-based translation, accounting for grammatical differences between Mandarin and Taiwanese Hokkien.

Unifying writing systems and Unicode encoding, we opt for Tai-lo as the representation method and develop a conversion process to unify corpora into Tai-lo format, mitigating inconsistencies across different writing systems and Unicode encoding issues.

IV. TRANSLATION MODEL AND TAIWANESE HOKKIEN LM

This section introduces our translation model based on the Transformer framework and two LM architectures for Taiwanese Hokkien: BERT and GPT-2.

A. Transformer Architecture for Translation Model

The Transformer is a sequence-to-sequence model utilizing self-attention. It comprises encoder and decoder blocks, each incorporating positional encoding to retain sequence order. The encoder utilizes multi-head attention (MHA) and feed forward network (FFN) layers, while the decoder introduces a masked MHA layer to prevent future context influence during decoding.

B. BERT and GPT-2 for Taiwanese Hokkien LMs

BERT, based on the Transformer encoder, offers bidirectional language understanding, enabling it to predict masked words from context during training. GPT-2, a unidirectional LM, utilizes the Transformer decoder with a masked MHA layer for generating coherent text, word by word, considering only preceding context for predictions.

V. RESULTS

In this section, we present the experimental results of our translation model and Taiwanese LMs, utilizing BERT and GPT-2 architectures. For the translation model, we evaluate its effectiveness in converting between Tai-lo and Full-han. The BLEU score serves as the evaluation metric for the translation model, while accuracy, F1 score, and perplexity are used for Taiwanese Hokkien LM evaluation.

A. Evaluation Methodologies

We employ diverse metrics for assessing both the translation model and Taiwanese Hokkien LM. For translation, BLEU score measures the similarity between model-generated and reference translations. For LMs, we use accuracy, F1 score, and perplexity to gauge performance. The training data comprises various texts, totaling about 43 million Chinese characters.

B. Results of Transformer-based Translation Model

We evaluate three sizes of Transformer architectures and a traditional RNN model for translation. While the RNN-

based model achieves the highest BLEU score, it suffers from longer translation times and repeated character errors. Hence, we select the Tiny Transformer as the preferred choice for its efficiency and relatively high performance.

C. Performance of Hokkien-BERT Taiwanese LM

Using the BERT model, we conduct experiments for two binary classification tasks: Bible text classification and news classification. The pre-trained Hokkien-BERT model demonstrates the capability to analyze word relationships in context, achieving robust performance in both tasks.

D. Performance of Hokkien-GPT Model for Taiwanese Text Generation

We train separate Hokkien-GPT models for Tai-lo and Full-han texts and evaluate their perplexity. The Full-han model exhibits ongoing improvement, while the Tai-lo model reaches its lowest perplexity at 8 hidden layers. Both models demonstrate impressive fluency in generating Taiwanese Hokkien text.

VI. CONCLUSIONS

This study underscores the significance of region-specific languages like Taiwanese Hokkien in natural language research. Despite challenges such as lacking an independent writing system, we developed a preprocessing pipeline to unify Taiwanese Hokkien text. Leveraging Transformer architecture, we trained conversion models between Tai-lo and Full-han writing systems. Our commitment extends to integrating Taiwanese Hokkien into AI and NLP. Through pretraining Hokkien-BERT and Hokkien-GPT models, we achieved excellent performance in downstream tasks, demonstrating their efficacy in capturing a universal language representation and generating fluent Taiwanese Hokkien text. These models lay the groundwork for further exploration of Taiwanese Hokkien within artificial intelligence.

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